Automated Construction of Multi-Element MEAM Interatomic Potentials

A DFT-informed neural-network optimization framework

code & data

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1 Alloy generator
random 12-element compositions — Python/NumPy

2 MD elastic database
$C_{11}, C_{12} \rightarrow E, \nu$ for 7 879 alloys — LAMMPS

5a Composition surrogate
composition → elastic moduli — TensorFlow

target C_{ij}

re-evaluate in LAMMPS, append, retrain

3 DFT references
a_0, E_{coh}, B, formation energies — Quantum Espresso

4 MEAM initialisation
Rose equation → α; cross-term signs — 12-element library

5b NN inverse optimisation
descend through the frozen surrogate to the target C_{ij}

Figure 1. The five-stage pipeline. The upper track assembles a composition–property database; the lower track builds the potential itself. Stage 5b closes the loop: optimized parameters are re-evaluated in LAMMPS, appended to the training set, and the surrogate is retrained.

1. Motivation & objective

The cubic stiffness constants C_{11}, C_{12} — and the derived Young's modulus E and Poisson's ratio ν — are the principal design criteria for structural alloys, yet measuring them is slow and expensive. Classical molecular dynamics recovers the same properties at negligible cost, *provided* an accurate interatomic potential is available. For metals the standard semi-empirical form is the Modified Embedded Atom Method (MEAM) [1, 2], whose parameters are traditionally fitted by hand — expert work that does not scale to many-element alloys.

Objective. Construct MEAM potentials automatically for an arbitrary set of elements, with no manual parameter fitting and no experimental input.

2. Random alloy generation

Compositions are drawn uniformly on the 12-element simplex (Al, Co, Cr, Cu, Fe, Mg, Mn, Mo, Ni, Si, Ti, Zn), weighted toward aluminium to oversample the commercially dominant region. Of ∼7 900 generated configurations, **7 815 are retained** by the physicality filter ($E > 0$, $0 < \nu < 0.48$, $C_{11} > C_{12}$).

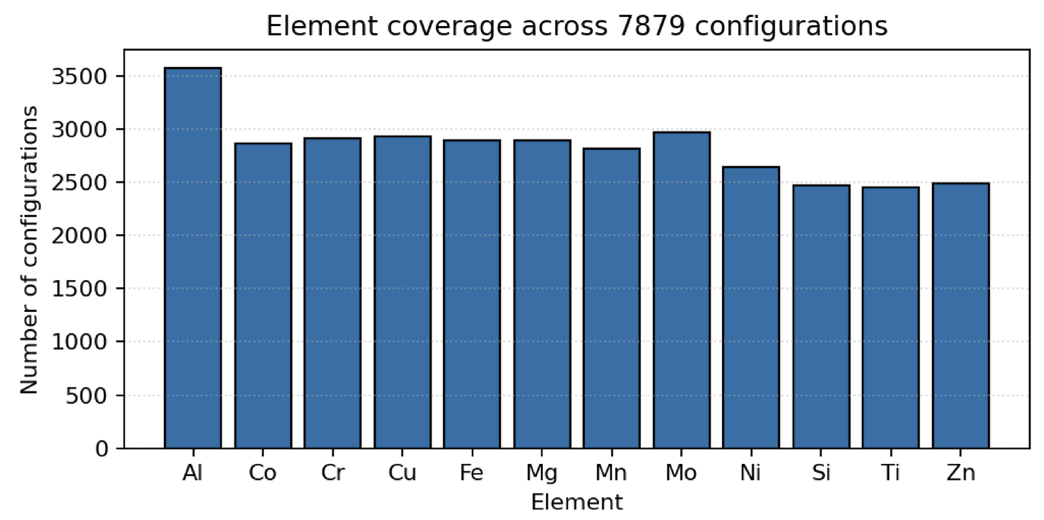

Figure 2. Element occurrence over the retained corpus.

3. Molecular-dynamics elastic database

Each configuration is realised as a ∼4 nm cubic supercell in LAMMPS [3] with the merged 12-element MEAM library. After anisotropic energy minimisation, axial strains $\delta = 10^{-3}$ along $\pm x, \pm y, \pm z$ yield C_{11} and C_{12} by central differences; E and $\nu = C_{12}/(C_{11} + C_{12})$ follow analytically. Median $E = 103.6$ GPa and median $\nu = 0.326$ over the corpus.

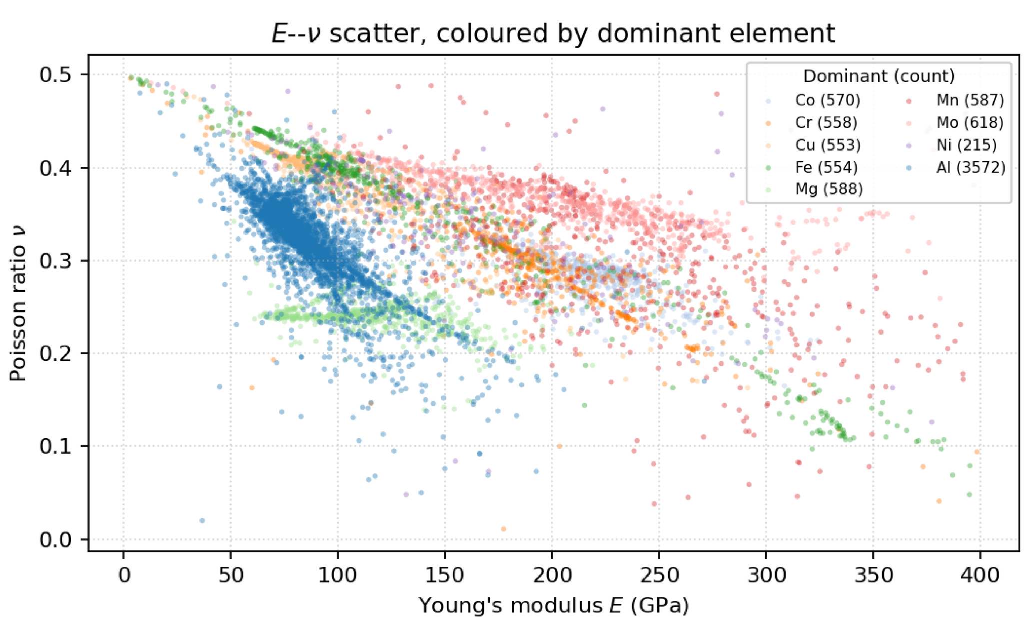

Figure 3. E–ν distribution coloured by dominant element ($N = 7\,815$). Named commercial alloys are highlighted.

4. DFT references & MEAM initialisation

Plane-wave PBE–GGA calculations in Quantum Espresso [4] provide each element's lattice constant a_0, cohesive energy E_{coh} and bulk modulus B from an equation-of-state fit. These pin the universal binding (Rose) curve [5] and fix the MEAM decay exponent

$$\alpha = \sqrt{9B\Omega/E_{coh}}, \qquad \Omega \text{ the atomic volume,}$$

ranging from $\alpha \approx 4.5$ (Al, Si) to 10.9 (Zn) across the twelve elements (FCC/BCC/HCP/diamond). Sixty-six binary formation energies seed the sign of each cross-term.

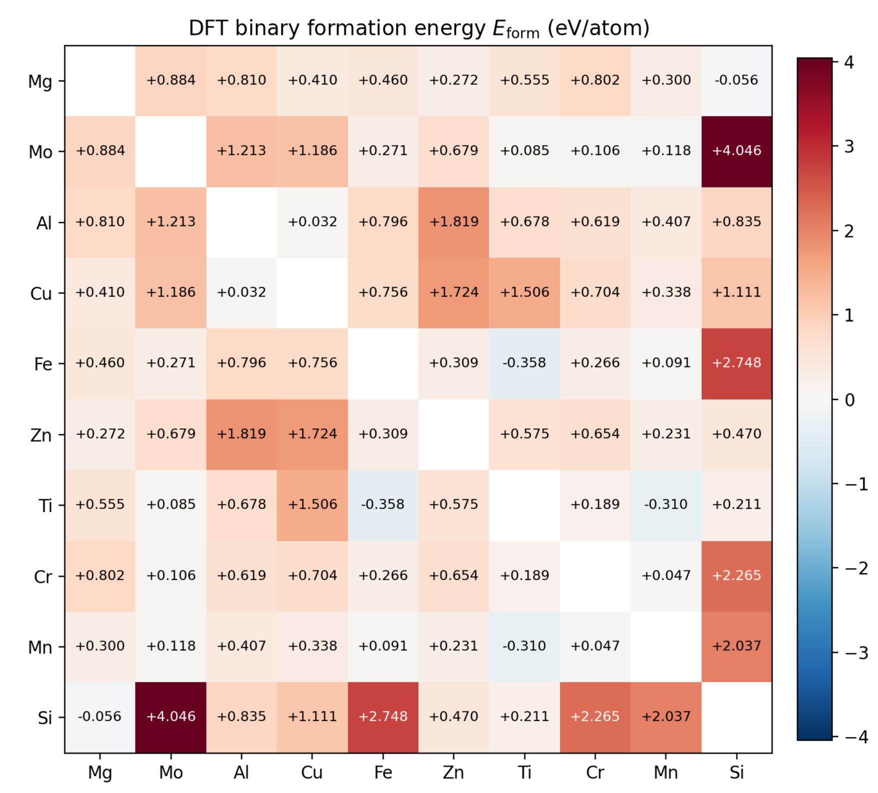

Figure 4. Binary formation energies $E_{form}(i, j)$ (eV/atom) from DFT; negative values indicate exothermic mixing.

5. Composition surrogate (stage 5a)

A fully connected $12 \rightarrow 32 \rightarrow 20 \rightarrow 2$ network (ReLU) predicts (C_{11}, C_{12}) under a Huber loss; E and ν are then recovered analytically, since direct ν regression is ill-conditioned. A deep ensemble of five seeds provides epistemic uncertainty.

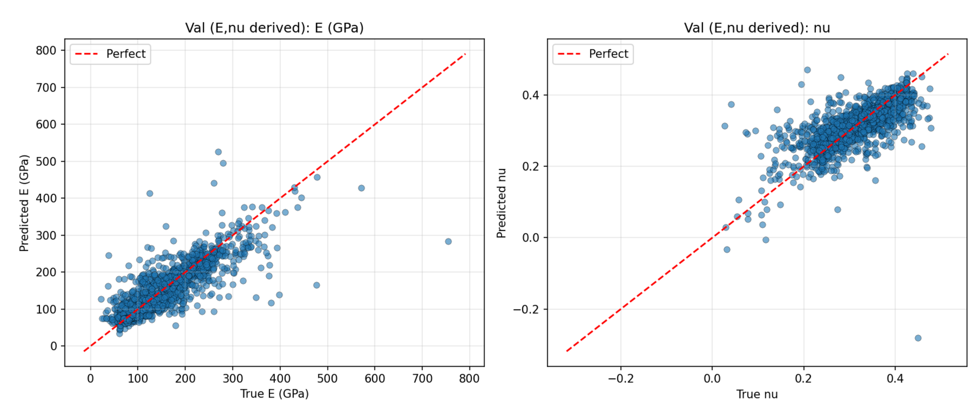

Figure 5. Parity between predicted and MD-computed moduli on the held-out validation set (30 % of the corpus).

$R^2_E = 0.77$, $R^2_\nu = 0.52$ on validation. For Al 7075 the surrogate predicts $E = 75.5$ GPa, $\nu = 0.334$, against published 71–72 GPa and $\nu \approx 0.33$ — within ∼5%.

6. NN-driven MEAM optimisation (stage 5b)

A second network learns the map (MEAM parameters) $\rightarrow (C_{11}, C_{12})$. It is fitted with Adam [6] on a Huber loss (which decays $1.0 \rightarrow 0.16$ in training); gradient descent through the *frozen* surrogate then locates the parameters that reproduce a target (C_{11}, C_{12}). Each optimum is re-evaluated in LAMMPS and appended to the training set, closing the active-learning loop.

Figure 6. Surrogate training: Huber loss versus epoch for the parameter-to-C_{ij} network.

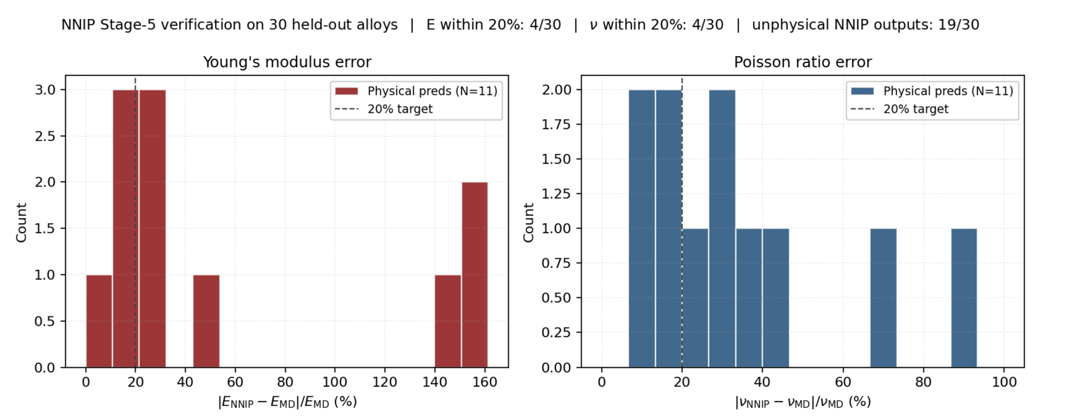

Figure 7. Verification of the optimized potential against the MD reference on unseen alloys.

Proof of concept. Of 30 unseen alloys, 11 remained physically stable under the optimized potential and 4 matched the MD reference within ∼20%. Accuracy grows with more data and longer training.

7. Validation against published data

		E MAPE (%)		ν MAPE (%)	
Family	N	ML	MEAM	ML	MEAM
Aluminium	33	**2.3**	7.4	4.0	4.5
Titanium	6	11.6	140.0	13.2	14.6
Stainless	3	69.6	324.1	75.5	47.3
Overall	43	**10.8**	42.2	**7.2**	7.1

Table 1. Mean absolute percentage error of the composition surrogate (ML) and the DFT-initialised potential (MEAM) against published elastic data, by alloy family.

Accuracy is strongest where the training corpus is densest (aluminium); sparse families inherit the corpus bias — a data limitation rather than a methodological one.

8. Conclusions & outlook

- An **automated, end-to-end 12-element MEAM pipeline**: from random compositions to an optimized multi-element potential with no hand fitting.
- **Aluminium-alloy elastic prediction** from composition alone, within a few percent of published values.
- **DFT-seeded, NN-trained MEAM**: first-principles references refit through a neural surrogate.

Future work: wider composition and crystal-structure coverage, longer surrogate training per refit, and denser DFT supercell sampling. The pipeline already runs end to end; what remains is more data, longer training, and sharper references.

References

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